Reproducible Data Analysis

George Dayoub

2026-04-13

1 Busiest Airports Analysis

1.1 Overview

Looking at both Table 1 and Figure 1 you can see that all the airports go up over time and the table helps show exactly the numbers of it. The graph makes it easier to see as the growth is visibly easier than looking at all the numbers.Starting from 2020 everything increases over time which makes sense because that was during the covid pandemic and travel was heavily restricted. Atlanta is the biggest the whole time represented it as it's the biggest and Dubai and Dallas and Tokyo are pretty close and also are growing. Frankfurt is lower than them with Beijing as well. From my goal it shows that Atlanta is the fastest and biggest and that Beijing and Frankfurt are lower, some numbers were not available to the Beijing which could affect it. But the main pattern is everything is going up and growing but some airports grow faster and have more people than others.

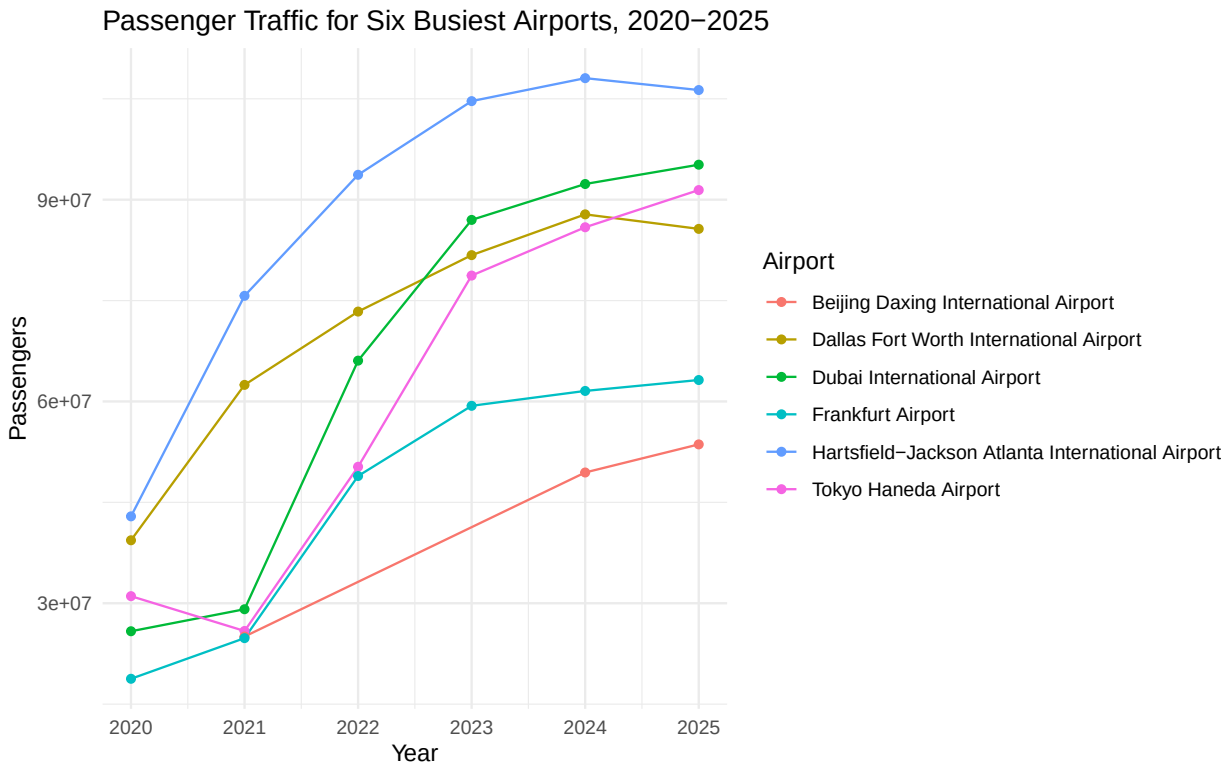
1.2 Table

Table 1: Passenger Traffic for Six Busiest Airports, 2020-2025

Airport	2025	2024	2023	2022	2021	2020
Beijing Daxing International Airport	53618949	49441029	NA	NA	25051012	NA
Dallas Fort Worth International Airport	85660127	87817864	81755538	73362946	62465756	39364990
Dubai International Airport	95200000	92331506	86994365	66069981	29110609	25836771
Frankfurt Airport	63189666	61564957	59355389	48918482	24812849	18768601
Hartsfield–Jackson Atlanta International Airport	106302208	108067766	104653451	93699630	75704760	42918685
Tokyo Haneda Airport	91430000	85900167	78719302	50290705	25876429	31055210

1.3 Plot

Figure 1: Passenger Traffic for Six Busiest Airports, 2020-2025



2 Monte Carlo Numerical Integration

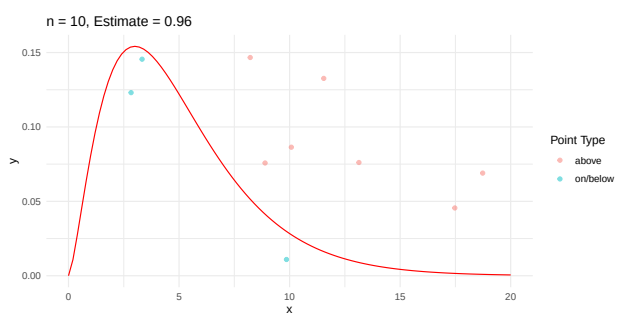
2.1 The Method

Monte Carlo integration is a way to estimate the area under a curve using random points. Instead of using a deterministic method like Riemann Sums you randomly sample points inside a rectangle that surrounds the function. You then check how many of those points fall on or below the curve and how many fall above it. The proportion of points that are on or below the curve multiplied by the area of the rectangle gives you an estimate of the integral. The more points you use the better and more stable the estimate gets.

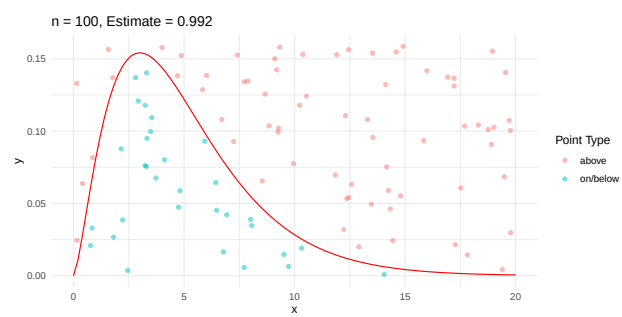
2.2 Visualization

Figure 2: Monte Carlo Numerical Integration of Chi-Square (df=5) at Four Resolutions

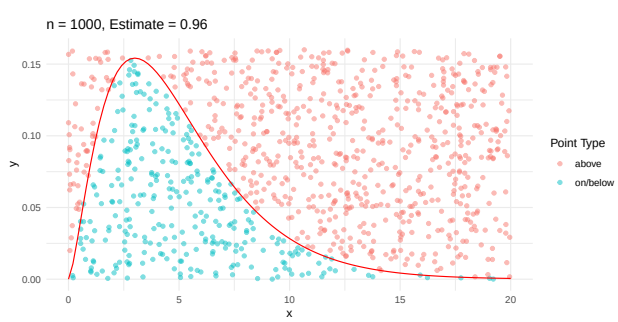
(a) n = 10



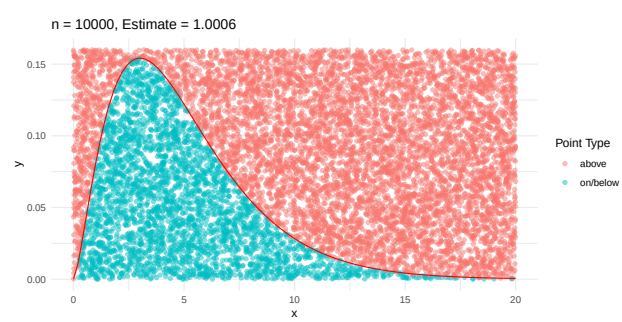
(b) n = 100



(c) n = 1000



(d) n = 10000



2.3 Analysis

The small multiple in Figure 2 shows how the estimate gets better as the number of points increases. When n = 10 and 100 the estimates are kinda off and jump around a lot but once you get to 1000 and 10000 they start to settle and stay close to the same value. The estimates in the last two plots are both really close to 1 and that tells me the actual value of the integral is about 1.

3 Planning and Prompting GenAI Tools

3.1 Plan

Goal:

- tidy the calcium data so it can be used to make a plot
- make a plot that shows how calcium levels change over time for both treatment groups so we can compare them

Needs:

Nouns: calcium.csv, treatment group, time point, calcium measurement, woman id, tidy data, plot

Verbs: load, pivot, combine, label, plot, compare

Steps:

1. load the calcium.csv file into R
2. the first 4 columns are the null group and the second 4 are the diet/exercise group so i need to separate them
3. add a treatment column to each group saying either null or diet/exercise
4. add a woman id column to keep track of each woman
5. pivot both groups longer so that the time points become one column and the measurements become one column
6. combine both groups into one tidy dataset
7. make a line plot or boxplot showing calcium levels over time with the two treatment groups shown in different colors
8. label everything so it is easy to read and compare

3.2 Plan-Informed Response

The following prompt was given to Anthropic Claude (Claude Sonnet 4.6) along with the calcium.csv file on April 13, 2026:

goal: tidy the calcium data in r and make a line plot comparing calcium levels over time between the two treatment groups. a few details to know:

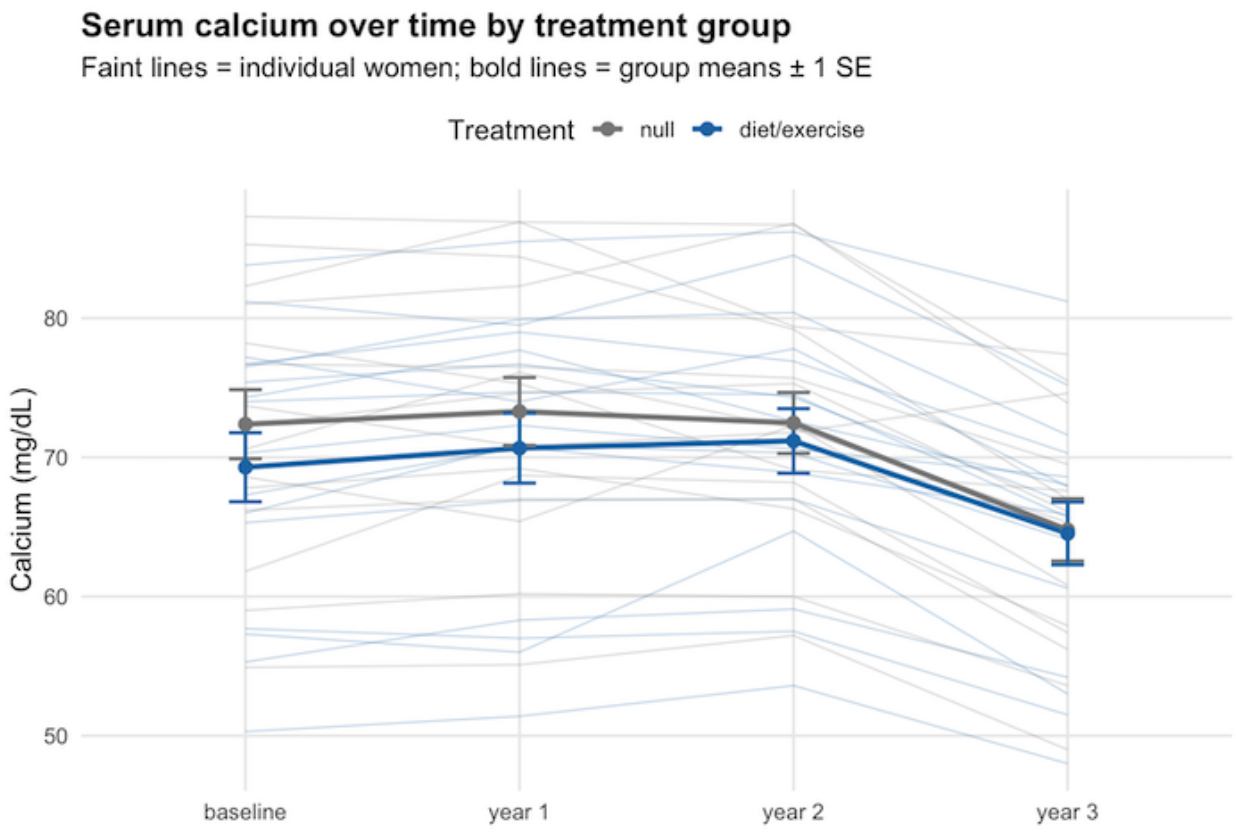
1. columns 1 to 4 = null group, columns 5 to 8 = diet/exercise group
2. time points are baseline, year 1, year 2, year 3
3. calcium is measured in mg/dL
4. load calcium.csv into r and give me the code
5. separate the two groups by column, and rename the columns in each to t1 to t4 so they line up
6. add a treatment column to each ("null" or "diet/exercise")
7. add a woman id column so each woman has a unique id across both groups
8. pivot both groups longer and be explicit about which columns to pivot so it doesn't accidentally grab the treatment column
9. combine into one tidy dataset and turn the time points into proper labels (baseline, year 1, year 2, year 3)
10. make a line plot of calcium over time, colored by treatment. show each woman as a faint line in the background, with bold mean lines and error bars on top

11. label everything clearly title subtitle units on the y-axis, clean theme. make it look nice

Claude returned the following R code and notes:

The following figures show the visualizations generated from the planned prompt.

Figure 3: Calcium Levels Over Time by Treatment Group - Plan Informed Response



```r
library(tidyverse)

# 1. Load (the file has duplicate column names y0..y3 repeated, so use name_repair)
raw <- read_csv("calcium.csv", name_repair = "minimal")

# 2. Split into the two groups by column position, rename to t1..t4
null_grp <- raw[, 1:4] |>
  set_names(c("t1", "t2", "t3", "t4"))

diet_grp <- raw[, 5:8] |>
  set_names(c("t1", "t2", "t3", "t4"))

# 3. Add treatment column to each
null_grp <- null_grp |> mutate(treatment = "null")
diet_grp <- diet_grp |> mutate(treatment = "diet/exercise")

# 4. Add a unique woman id across both groups
null_grp <- null_grp |> mutate(woman_id = row_number())
diet_grp <- diet_grp |> mutate(woman_id = row_number() + nrow(null_grp))

# 5. Pivot longer - explicitly name t1:t4 so treatment/woman_id stay put
null_long <- null_grp |>
  pivot_longer(cols = t1:t4, names_to = "time", values_to = "calcium")

diet_long <- diet_grp |>
  pivot_longer(cols = t1:t4, names_to = "time", values_to = "calcium")

# 6. Combine + relabel time points, drop the empty row
calcium_tidy <- bind_rows(null_long, diet_long) |>
  drop_na(calcium) |>
  mutate(
    time = recode(time,
                  t1 = "baseline",
                  t2 = "year 1",
                  t3 = "year 2",
                  t4 = "year 3"),
    time = factor(time, levels = c("baseline", "year 1", "year 2", "year 3")),
    treatment = factor(treatment, levels = c("null", "diet/exercise"))
  )

# 7. Summary stats for the bold mean lines + error bars (SE)
calcium_summary <- calcium_tidy |>
  group_by(treatment, time) |>
  summarise(
    mean_ca = mean(calcium),
    se      = sd(calcium) / sqrt(n()),
    .groups = "drop"
  )

# 8. Plot
ggplot(calcium_tidy, aes(x = time, y = calcium,
                         color = treatment, group = treatment)) +
  geom_line(aes(group = woman_id), alpha = 0.25, linewidth = 0.4) +
  geom_line(data = calcium_summary, aes(y = mean_ca),
```

```r
            linewidth = 1.2) +
  geom_errorbar(data = calcium_summary,
                aes(y = mean_ca,
                    ymin = mean_ca - se,
                    ymax = mean_ca + se),
                width = 0.12, linewidth = 0.8) +
  geom_point(data = calcium_summary, aes(y = mean_ca),
             size = 2.5) +
  scale_color_manual(values = c("null" = "#888888",
                                "diet/exercise" = "#1f77b4")) +
  labs(
    title    = "Serum calcium over time by treatment group",
    subtitle = "Faint lines = individual women; bold lines = group means ± 1 SE",
    x        = NULL,
    y        = "Calcium (mg/dL)",
    color    = "Treatment"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "top",
    plot.title      = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )
```

Claude also noted that the CSV has duplicated column names so `name_repair = "minimal"` was used, woman IDs are unique across both groups, and error bars show mean ± 1 SE.

3.3 Generic Response

The generic prompt used was: "Create a data visualization that will help me understand these data. Calcium.csv"

Claude's response to the generic prompt was as follows: Both groups follow the same pattern — values stay relatively stable across the first three waves (y0 to y1 to y2), then drop noticeably at the final measurement (y3). Group A starts slightly higher (baseline mean ~72.4 vs. ~69.3 for Group B), but both groups converge to nearly identical final values (~64.8 and ~64.5). There is substantial individual variation as well.

The following figures show the visualizations generated from the generic prompt.

Figure 4: Group Means View of Calcium Data

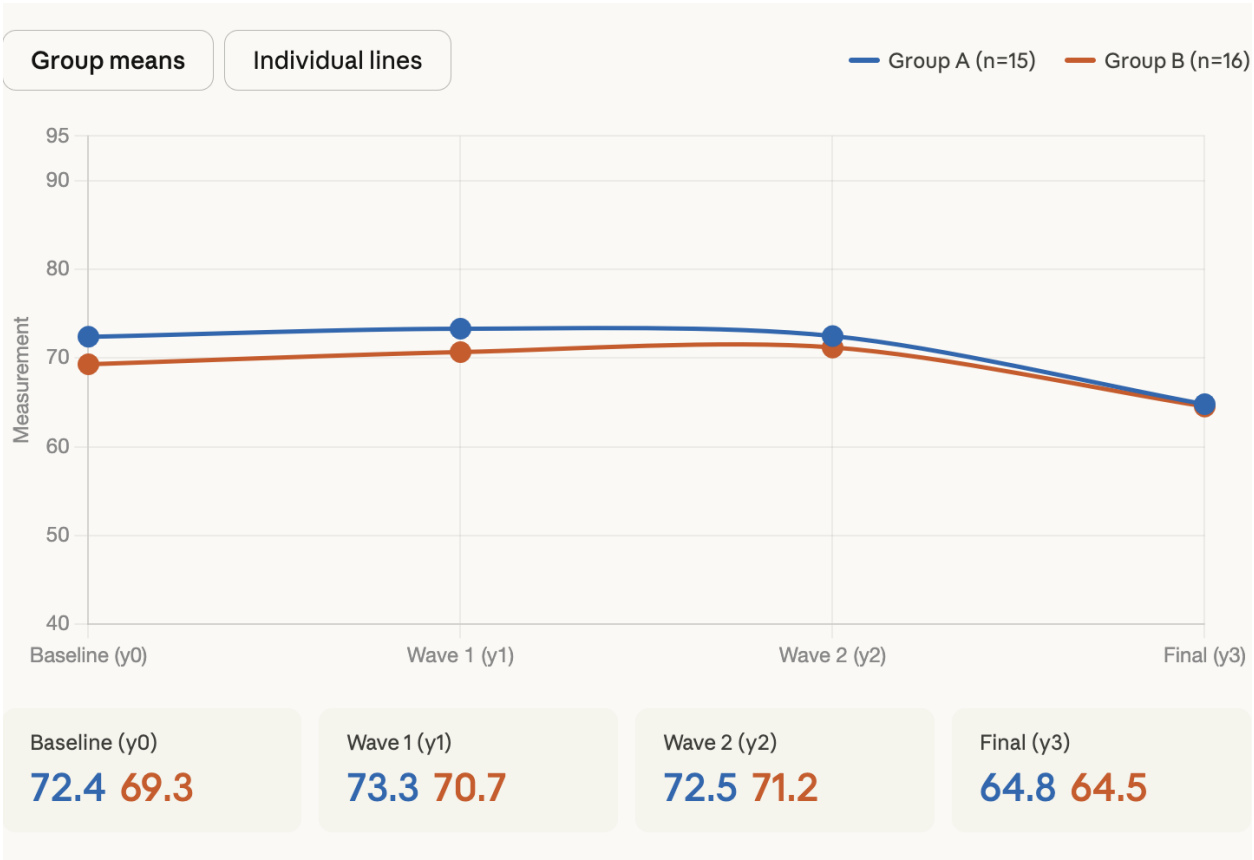
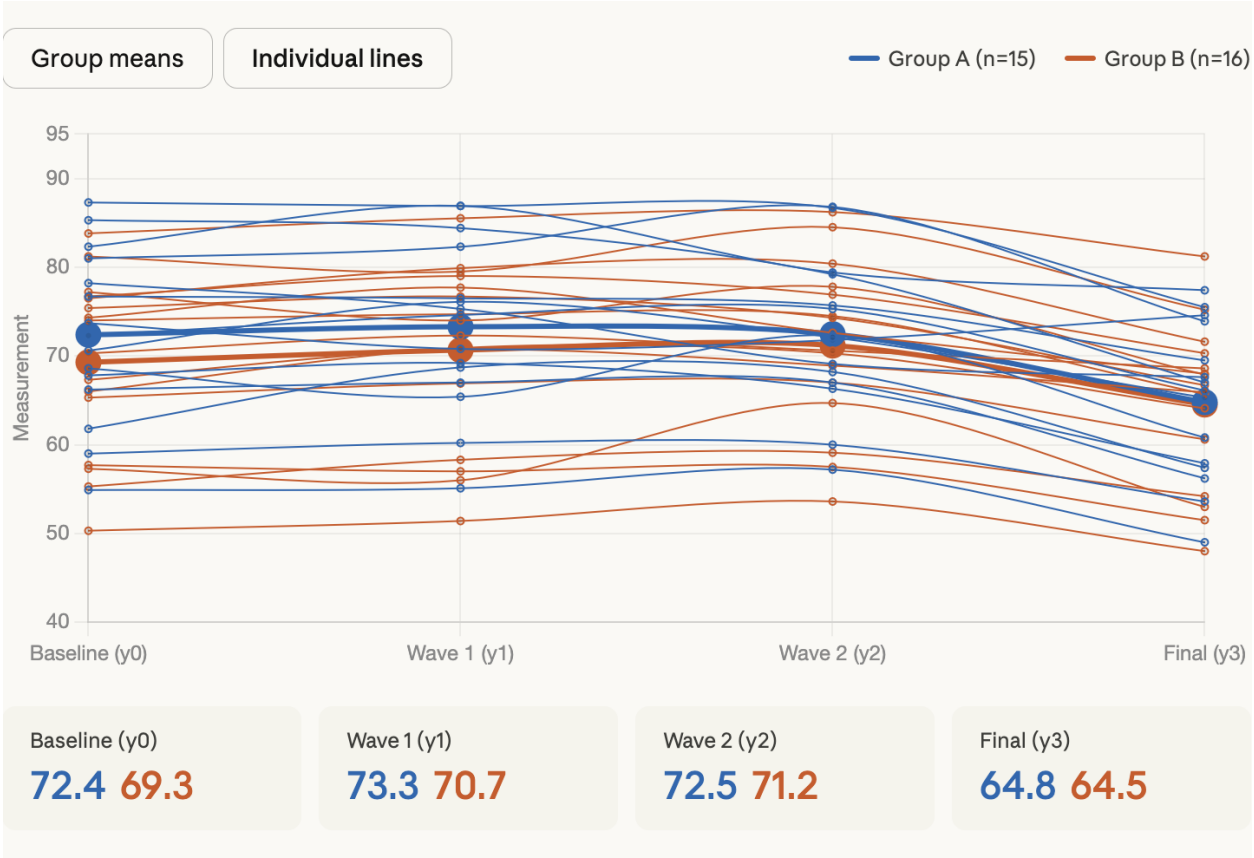


Figure 5: Individual Lines View of Calcium Data



3.4 Comparison

Looking at both responses you can tell that giving Claude a plan made a big difference. When it was given the generic prompt it just made a plot but it didn't know what the groups were or what the columns meant so it just called them group a and group b which doesn't really tell you anything. When I gave it my plan informed prompt, it actually knew that the first 4 columns were the null group and the last 4 were the diet/exercise group and it labeled everything correctly. The code it gave me with the plan was also way more detailed and specific to the actual data like it knew to pivot longer and add a treatment column and a woman id. The generic one just kind of guessed at the structure. So the main lesson is that if you give GenAI a good plan with details about the data it does a way better job than if you just throw the file at it and hope for the best.

4 Reflection

So far in this course I feel like I learned a lot more than I expected. Before this class I kind of just wrote code and hoped it worked but now I actually plan things out first before I start coding which makes everything way easier. For the airports assignment I had to think about what the data needed to look like before I could even write the wrangling code and that made the whole thing go smoother.

I also learned a lot about making plots look good and actually mean something. Before I would just throw data into a plot and call it done but now I think about what geometry to use, what the labels should say, and whether the plot actually helps someone understand the data. The pirates assignment helped me see that a scatter plot can show a pattern even when the relationship is weak.

Coming into the GenAI section I already knew that a detailed prompt would lead to better results than a generic one because the model has more context to base off of. So it was not that surprising to me that the plan informed response was better but it was still useful to see it side by side with the generic one because it showed exactly what kind of context matters most like knowing what the columns meant and what the groups were.

The Quarto file and the structure was new to me too and I see how use it is because everything is in one place. Having the code and the writeup being connected so someone else could actually reproduce what I did which was something I never thought about before this class. While working on this I realized the importance of a quarto document, as it allows the reader to follow along giving the author structure to layout their research.

5 Author Contributions

George Dayoub: Formal Analysis, Visualization, Writing – Original Draft, Writing – Review & Editing.

6 GenAI Usage Appendix

6.1 Usage 1: Plan-Informed Calcium Prompt

Tool: Anthropic Claude (Claude Sonnet 4.6)

Date: April 13, 2026

Prompt:

goal: tidy the calcium data in r and make a line plot comparing calcium levels over time between the two treatment groups. a few details to know:

1. columns 1 to 4 = null group, columns 5 to 8 = diet/exercise group
2. time points are baseline, year 1, year 2, year 3
3. calcium is measured in mg/dL
4. load calcium.csv into r and give me the code
5. separate the two groups by column, and rename the columns in each to t1 to t4 so they line up
6. add a treatment column to each ("null" or "diet/exercise")
7. add a woman id column so each woman has a unique id across both groups
8. pivot both groups longer and be explicit about which columns to pivot so it doesn't accidentally grab the treatment column
9. combine into one tidy dataset and turn the time points into proper labels (baseline, year 1, year 2, year 3)
10. make a line plot of calcium over time, colored by treatment. show each woman as a faint line in the background, with bold mean lines and error bars on top
11. label everything clearly title subtitle units on the y-axis, clean theme. make it look nice

Response: Claude returned the R code shown in the Plan-Informed Response section along with notes that the CSV has duplicated column names so `name_repair = "minimal"` was used, woman IDs are unique across both groups, and error bars show mean ± 1 SE.

6.2 Usage 2: Generic Prompt (Provided by Assignment)

Tool: Anthropic Claude (Claude Sonnet 4.6 Extended)

Date: March 31, 2026

Prompt: *"Create a data visualization that will help me understand these data. Calcium.csv"*

Response: Both groups follow the same pattern — values stay relatively stable across the first three waves (y0 to y1 to y2), then drop noticeably at the final measurement (y3). Group A starts slightly higher (baseline mean ~72.4 vs. ~69.3 for Group B), but both groups converge to nearly identical final values (~64.8 and ~64.5). There is substantial individual variation as well. Claude also produced two interactive visualizations showing group means and individual lines.

7 Code Appendix

```
# Load Packages -
## Author: George Dayoub

library(rvest)
library(dplyr)
library(tidyr)
library(knitr)
library(kableExtra)
library(ggplot2)
library(tidyverse)
## Author: George Dayoub

# read the wikipedia page
url <-
 ↪  "https://en.wikipedia.org/wiki/List_of_busiest_airports_by_passenger_traffic"
page <- read_html(url)

# only grab the wikipedia data tables
tables <- page %>%
  html_elements("table.wikitable") %>%
  html_table(fill = TRUE)

# grab the yearly tables and add the year
t2025 <- tables[[1]] %>% mutate(year = 2025)
t2024 <- tables[[2]] %>% mutate(year = 2024)
t2023 <- tables[[3]] %>% mutate(year = 2023)
t2022 <- tables[[4]] %>% mutate(year = 2022)
t2021 <- tables[[5]] %>% mutate(year = 2021)
t2020 <- tables[[6]] %>% mutate(year = 2020)

# combine all years into one dataset
airports <- bind_rows(t2025, t2024, t2023, t2022, t2021, t2020)

# clean and organize the data
airports_tidy <- airports %>%
  filter(!is.na(Airport)) %>%   # remove empty rows
  select(Airport, Totalpassengers, year) %>%   # keep only needed columns
  rename(passengers = Totalpassengers) %>%     # rename passenger column
  mutate(
    passengers = gsub(",", "", passengers),         # remove commas
    passengers = gsub("\\[.*\\]", "", passengers),  # remove citation numbers
     ↪  like the random number in [19]
    passengers = as.numeric(passengers)             # turn text into numbers
  ) %>%
  filter(Airport %in% c(
```

```r
    "Hartsfield-Jackson Atlanta International Airport",
    "Frankfurt Airport",
    "Beijing Daxing International Airport",
    "Dubai International Airport",
    "Tokyo Haneda Airport",
    "Dallas Fort Worth International Airport"
  ))
## Author: George Dayoub

# use tidy airport data from the last question
airports_table <- airports_tidy %>%

# spread the year values across columns
  pivot_wider(
    names_from = year,      #make each year its own column
    values_from = passengers # those columns with passenger values
  ) %>%

# sort from a-z by airport name
  arrange(Airport)

# use kable to print the table nicely
kable(airports_table)
## Author: George Dayoub

# start with the tidy airport data from the last question
airports_tidy %>%
  # set up the graph
  ggplot(aes(x = year, y = passengers, color = Airport, group = Airport)) +

# add lines so i can see how each airport changes over time
  geom_line() +  # geom_line for line plot

# add points for each airport in each year
  geom_point() +
  # add labels so it makes sense
  labs(
    title = "Passenger Traffic for Six Busiest Airports, 2020-2025",
    x = "Year",
    y = "Passengers",
    color = "Airport"
  ) +
  scale_x_continuous(breaks = c(2020, 2021, 2022, 2023, 2024, 2025)) +

  # make it look cleaner like what i did for first 2 questions
  theme_minimal()
## Author: George Dayoub
```

```r
#' MonteCarlo sim function
#'
#' his function runs a Montecarlo simulation by generating random points
#' within given x and y bounds and estimating the area under a function
#'
#' @param n number of random points
#' @param xmin minimum x value
#' @param xmax maximum x value
#' @param ymin minimum y value
#' @param ymax maximum y value
#'
#' @return a data frame with x, y, function values, flags, and estimate
#' @examples
#' mc_simulation(1000, 2, 6, 0, 4)

# name
mc_simulation <- function(n, xmin, xmax, ymin, ymax) {

  # n = how many random points i want
  # xmin and xmax = x bounds
  # ymin and ymax = y bounds

  x <- runif(n, min = xmin, max = xmax)  # make random x values in the x interval
  y <- runif(n, min = ymin, max = ymax)  # make random y values in the y interval

  # define the function
  fx <- -(x - 4)^2 + 4

  # rectangle area
  rect_area <- (xmax - xmin) * (ymax - ymin)

  # flag each point
  flag <- ifelse(y <= fx, "on/below", "above")

  # find the proportion of points on or below the curve
  prop_under <- mean(flag == "on/below")

  # estimate the integral
  estimate <- rect_area * prop_under

  # combine into one data frame
  points_df <- data.frame(
    x = x,
    y = y,
    fx = fx,
    flag = flag,
    estimate = estimate
  )
```

```r
  return(points_df)
}
## Author: George Dayoub

# calculates the rectangle area
rect_area <- (20 - 0) * (0.16 - 0)

# n = 10
points10 <- mc_simulation(10, 0, 20, 0, 0.16)
points10 <- points10 %>%
  mutate(
    fx = dchisq(x, df = 5),  # plug x into the correct function
    flag = ifelse(y <= fx, "on/below", "above")  # check if point is under the
     ↪  curve
  )
# get the estimate for n = 10
estimate10 <- rect_area * mean(points10$flag == "on/below")

# n = 100
points100 <- mc_simulation(100, 0, 20, 0, 0.16)
points100 <- points100 %>%
  mutate(
    fx = dchisq(x, df = 5),  # plug x into the correct function
    flag = ifelse(y <= fx, "on/below", "above")  # check if point is under the
     ↪  curve
  )
# get the estimate for n = 100
estimate100 <- rect_area * mean(points100$flag == "on/below")

# n = 1000
points1000 <- mc_simulation(1000, 0, 20, 0, 0.16)
points1000 <- points1000 %>%
  mutate(
    fx = dchisq(x, df = 5),  # plug x into the correct function
    flag = ifelse(y <= fx, "on/below", "above")  # check if point is under the
     ↪  curve
  )
# get the estimate for n = 1000
estimate1000 <- rect_area * mean(points1000$flag == "on/below")

# n = 10000
points10000 <- mc_simulation(10000, 0, 20, 0, 0.16)
points10000 <- points10000 %>%
  mutate(
    fx = dchisq(x, df = 5),  # plug x into the correct function
    flag = ifelse(y <= fx, "on/below", "above")  # check if point is under the
     ↪  curve
```

```r
  )
# get the estimate for n = 10000
estimate10000 <- rect_area * mean(points10000$flag == "on/below")
## Author: George Dayoub

# plots everything so i can actually see what's happening

# n = 10
ggplot(points10, aes(x = x, y = y, color = flag)) +
  geom_point(alpha = 0.5) +  # plot random points
  stat_function(fun = dchisq, args = list(df = 5), xlim = c(0, 20), color =
   ↪  "red") +  # plot the actual function
  labs(
    title = paste("n = 10, Estimate =", round(estimate10, 4)),
    x = "x",
    y = "y",
    color = "Point Type"
  ) +
  theme_minimal()

# n = 100
ggplot(points100, aes(x = x, y = y, color = flag)) +
  geom_point(alpha = 0.5) +  # plot random points
  stat_function(fun = dchisq, args = list(df = 5), xlim = c(0, 20), color =
   ↪  "red") +  # plot the actual function
  labs(
    title = paste("n = 100, Estimate =", round(estimate100, 4)),
    x = "x",
    y = "y",
    color = "Point Type"
  ) +
  theme_minimal()

# n = 1000
ggplot(points1000, aes(x = x, y = y, color = flag)) +
  geom_point(alpha = 0.5) +  # plot random points
  stat_function(fun = dchisq, args = list(df = 5), xlim = c(0, 20), color =
   ↪  "red") +  # plot the actual function
  labs(
    title = paste("n = 1000, Estimate =", round(estimate1000, 4)),
    x = "x",
    y = "y",
    color = "Point Type"
  ) +
  theme_minimal()

# n = 10000
ggplot(points10000, aes(x = x, y = y, color = flag)) +
```

```r
geom_point(alpha = 0.5) +  # plot random points
stat_function(fun = dchisq, args = list(df = 5), xlim = c(0, 20), color =
 ↪  "red") +  # plot the actual function
labs(
  title = paste("n = 10000, Estimate =", round(estimate10000, 4)),
  x = "x",
  y = "y",
  color = "Point Type"
) +
theme_minimal()
```